Jim Ngo

Ontario, CA | (909) 830-2264 | jim.ngo@outlook.com

jimsngo.github.io
linkedin.com/in/jimsngo

PROFESSIONAL SUMMARY

Electrical Engineer & Senior Electronics Technician with extensive experience in hardware testing, R&D prototyping, SMPS modeling (LTspice), and RF systems. Proven track record in prototype fabrication, PCB debugging, and electro-mechanical assembly within medical, defense, and industrial computing environments.

TECHNICAL SKILLS

- **Testing & Hardware:** Computer Hardware Validation, Medical Surgical Consoles, Subsystem Verification, ICT & Functional Testing, Analog/Digital/Microcontroller Circuits, SMT Soldering.
- **Power & RF:** Power Supply (SMPS) Modeling, Worst-Case Analysis, RF Testing (Up/Downconverters, Tx/Rx), Custom Inductors & Transformers.
- **Tools & Languages:** Logic Analyzers, High-Speed Oscilloscopes, LTspice, Verilog, C, Python, LabVIEW, MATLAB, Windows/Linux Scripting, HTML/CSS.

WORK EXPERIENCE

Alcon | *Senior Electronic Lab Technician* | Lake Forest, CA | 03/2022 – 06/2022

- Fabricated prototype instrumentation and conducted V&V on surgical consoles, subsystems, and PCBs using logic analyzers and oscilloscopes.
- Isolated and resolved system-level hardware anomalies alongside Senior R&D Engineers; performed precision electro-mechanical assembly.

General Micro Systems | *Engineering R&D Technician* | Rancho Cucamonga, CA | 02/2020 – 03/2022

- Built, tested, debugged, and validated ruggedized computing hardware, network switches, and routers for R&D and Manufacturing.
- Authored test procedures and assembly instructions; deployed custom Linux/Windows scripts for automated hardware validation.

Micro Analog Inc. | *Senior Electronic Technician* | La Verne, CA | 10/2019 – 02/2020

- Executed ICT and functional testing on complex analog, digital, and microcontroller boards; diagnosed root-cause component failures and performed SMT rework.

Business Ownership & Financial Services | *Franchise Owner* | 10/1990 – 10/1999

- Managed real estate/mortgage brokerage (2015–2019; BRE 01164334 | NMLS 1353214), AAMCO Transmissions (1995–2015), and Firestone Tires (1990–1995).

Hughes Aircraft Company | *Electrical Engineer* | El Segundo, CA | 12/1985 – 10/1990

- Designed/tested Low Voltage Power Supplies (LVPS); performed worst-case efficiency/stability analyses and aligned RF communication equipment.

EDUCATION

- **MSEE** – Loyola Marymount University, Los Angeles, CA
- **BSEE** – California State Polytechnic University, Pomona, CA